

Word and Concept Embeddings

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Logistics

- (1) Review session on April 13
- (2) In-class final on April 15
- (3) Coding assignment 3 will be posted ASAP (likely this evening) and is due on May 2

Last time: “bag of words” models

- (+) Count-based approaches are simple and work surprisingly well in practice
- (+) Often the best approach with small datasets
- (-) Word representations are highly sparse and therefore inefficient
- (-) Does not capture word order or interpret words in context
- (-) Does not group synonyms together or understand semantic relationships between words



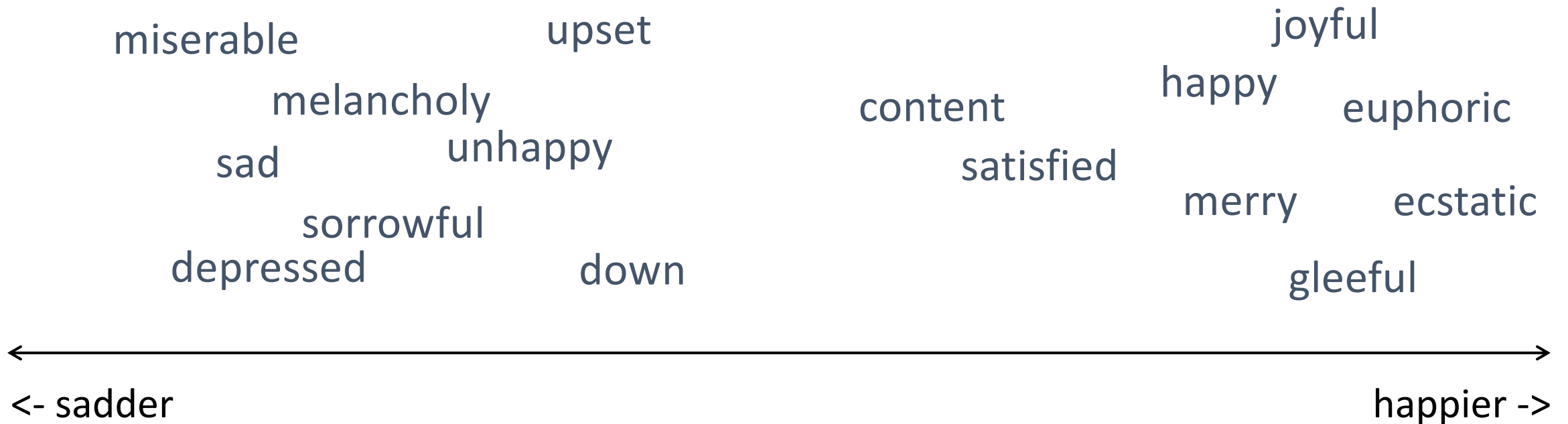
Today

- Motivating word embeddings
- How we learn word embeddings
- What we can do now that we have word embeddings
 - Embedding-based models
 - Word/concept retrieval
- Medical concept embeddings

Motivating word embeddings

Key problem: quantifying semantic relationships between words

Word embedding: a numeric representation of a word that encodes its meaning



Numeric value indicating whether the word is happy or sad

Training a robot to buy groceries



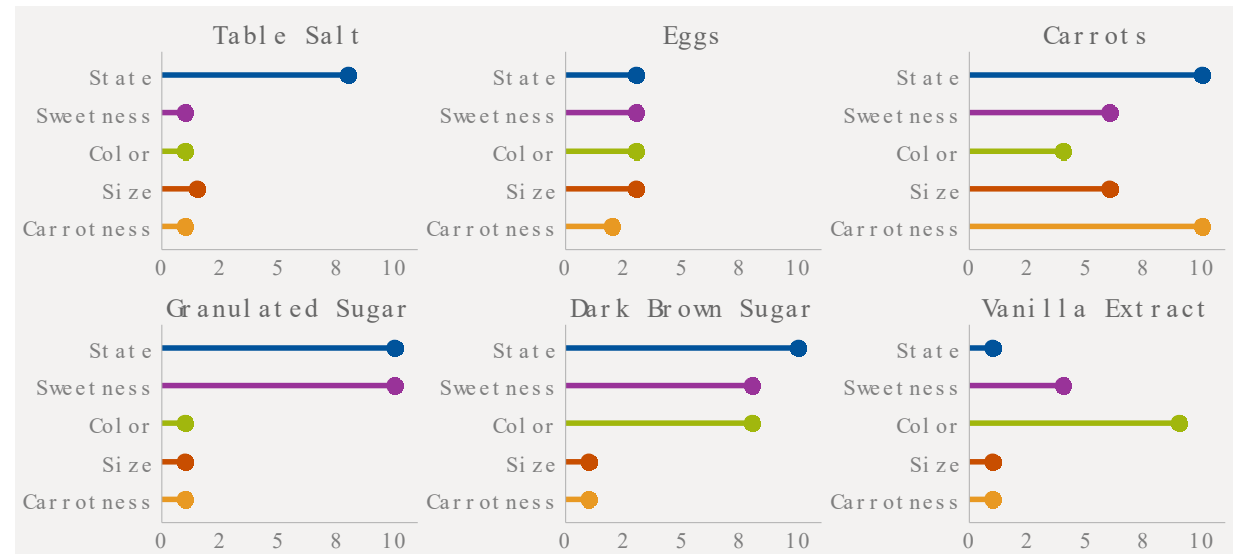
Example from Anand Chowdhury, MMCI 2019

Grocery List

- ☐ granulated sugar
- ☐ vanilla extract
- ☐ dark brown sugar
- ☐ carrots
- ☐ table salt
- ☐ eggs

Identify items by their attributes (including previously unseen items)

Dimension	1	10
State	Liquid	Solid
Sweetness	Bland	Sweet
Color	Light	Dark
Size	Small	Large
Carrotiness	Not really	Platonic essence of carrot



Why does this help us?

- The model can make sense of words it hasn't seen before (weren't used in training)
- Similar words (e.g. synonyms) will have similar attributes, and therefore will have similar effect on model predictions
- (more complicated) Now we can convert text to a sequence of vectors; and we were already very good at making predictions from sequences of vectors

How do we learn word embeddings?

Key idea: words are defined by the contexts in which they appear

How do we learn these attributes?

KEY IDEA: words are *defined* by the contexts in which they appear

A **man** strolls down the street

A **woman** strolls down the street

A **child** strolls down the street

A **crocodile** strolls down the street

A **banana** strolls down the street

A **concept** strolls down the street

Consider substituting one word for another.

Claim: if two words are equally likely to appear all possible contexts, then the two words mean the same thing.

Step 1: define *co-occurrence* or *context*

{input word, context word}

{strolls, man}

{strolls, woman}

{swims, crocodile}

{swims, fish}

{flies, bird}

{flies, plane}

We define a context word as one that appears inside a fixed-length window around the input word in our training corpus.

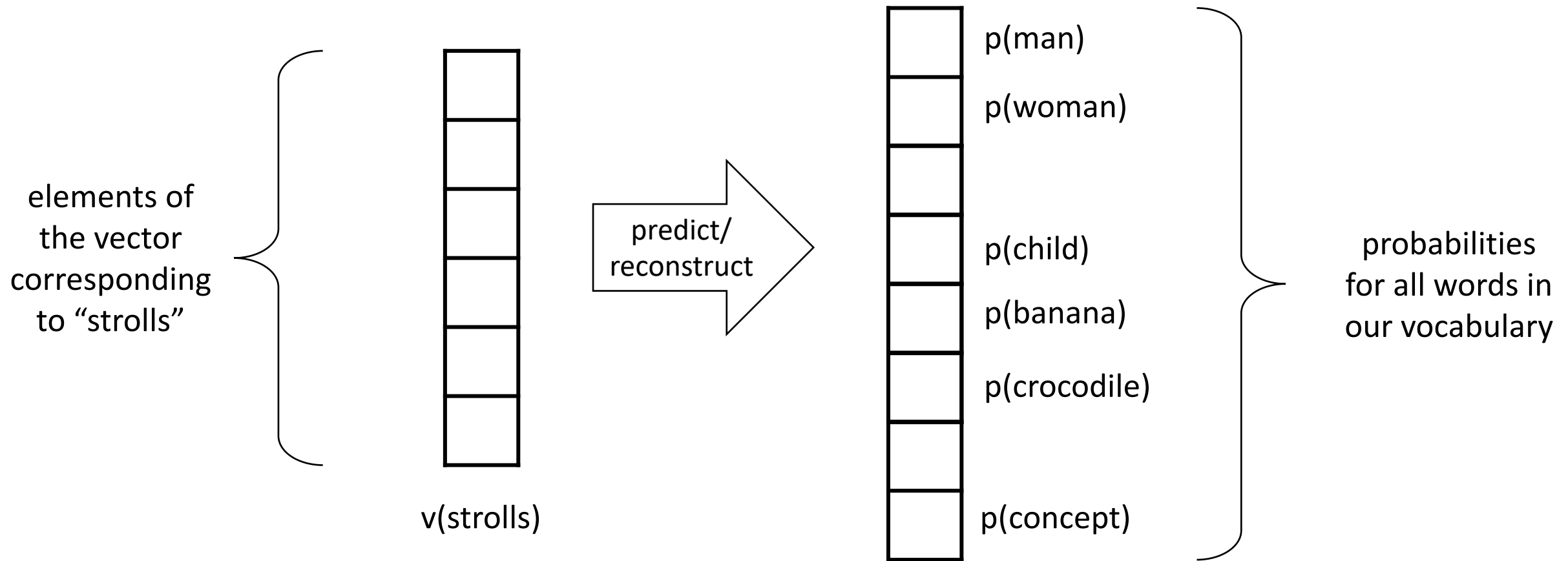
(e.g. Wikipedia)

A man strolls down the street.

input context

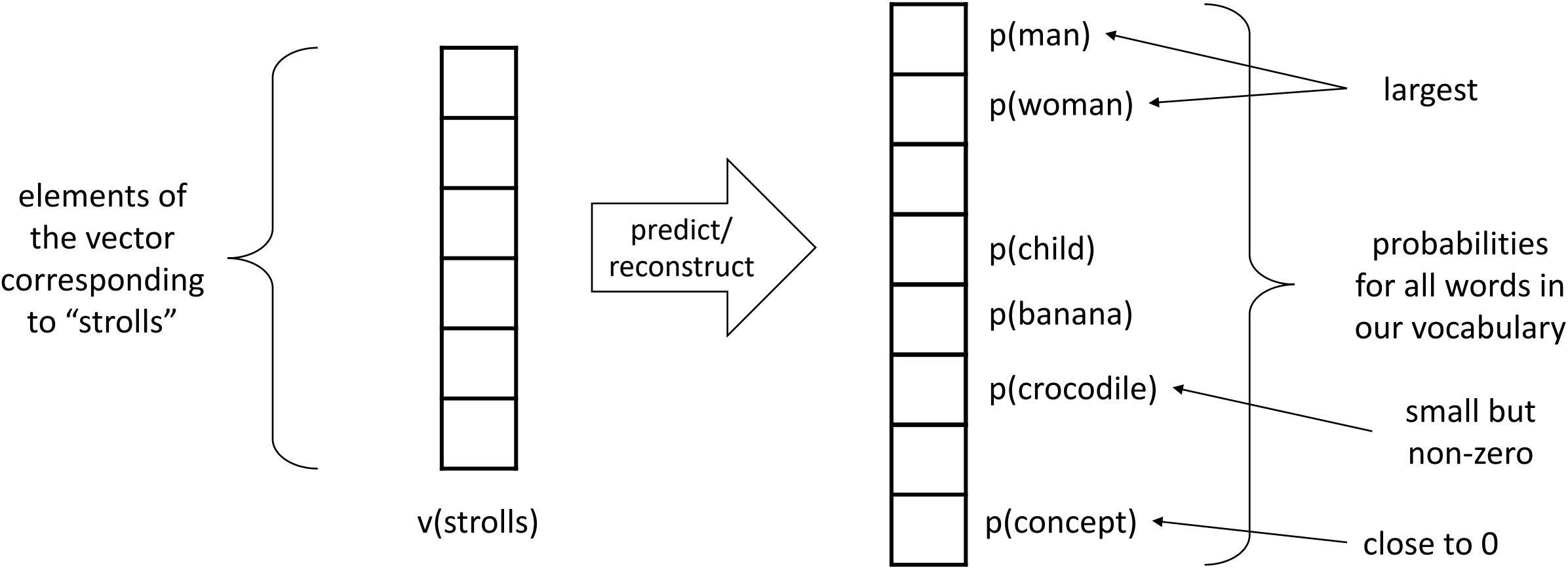
Step 2: find vector representations sufficient
to predict the likely context

(i.e. what other words are likely/unlikely to be nearby)



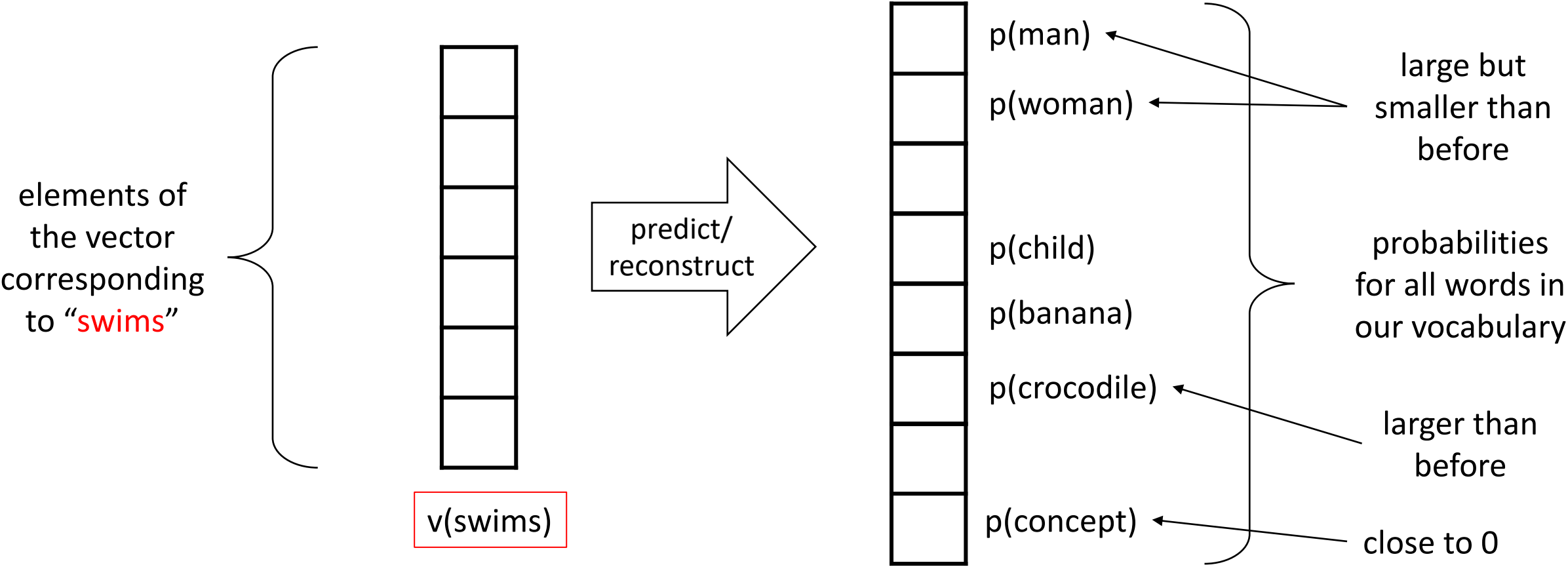
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Step 2: find vector representations sufficient to predict the likely context

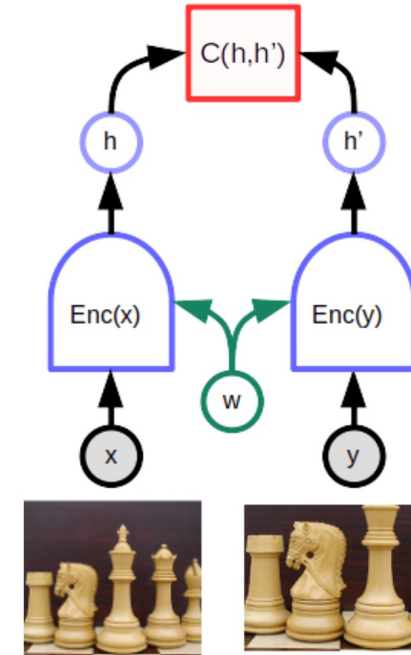
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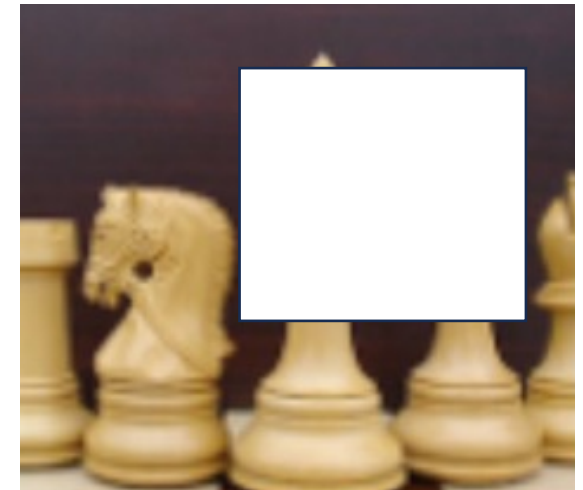
Recall: self-supervised learning

- We are predicting context words from input words (skip-gram)
- Later we will consider a noise contrastive version of this

contrastive /
noise
contrastive

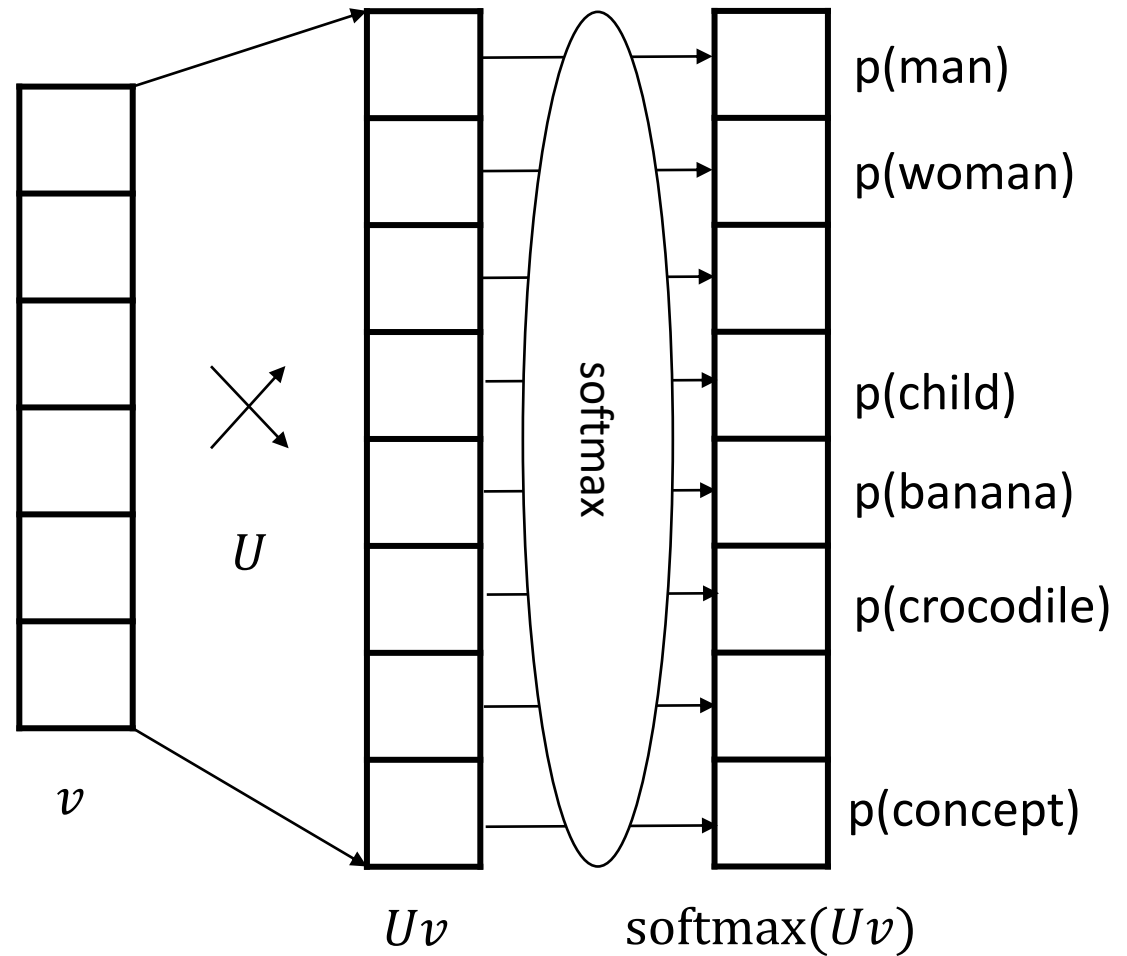


masked token /
patch / sequence
prediction



The skip-gram model

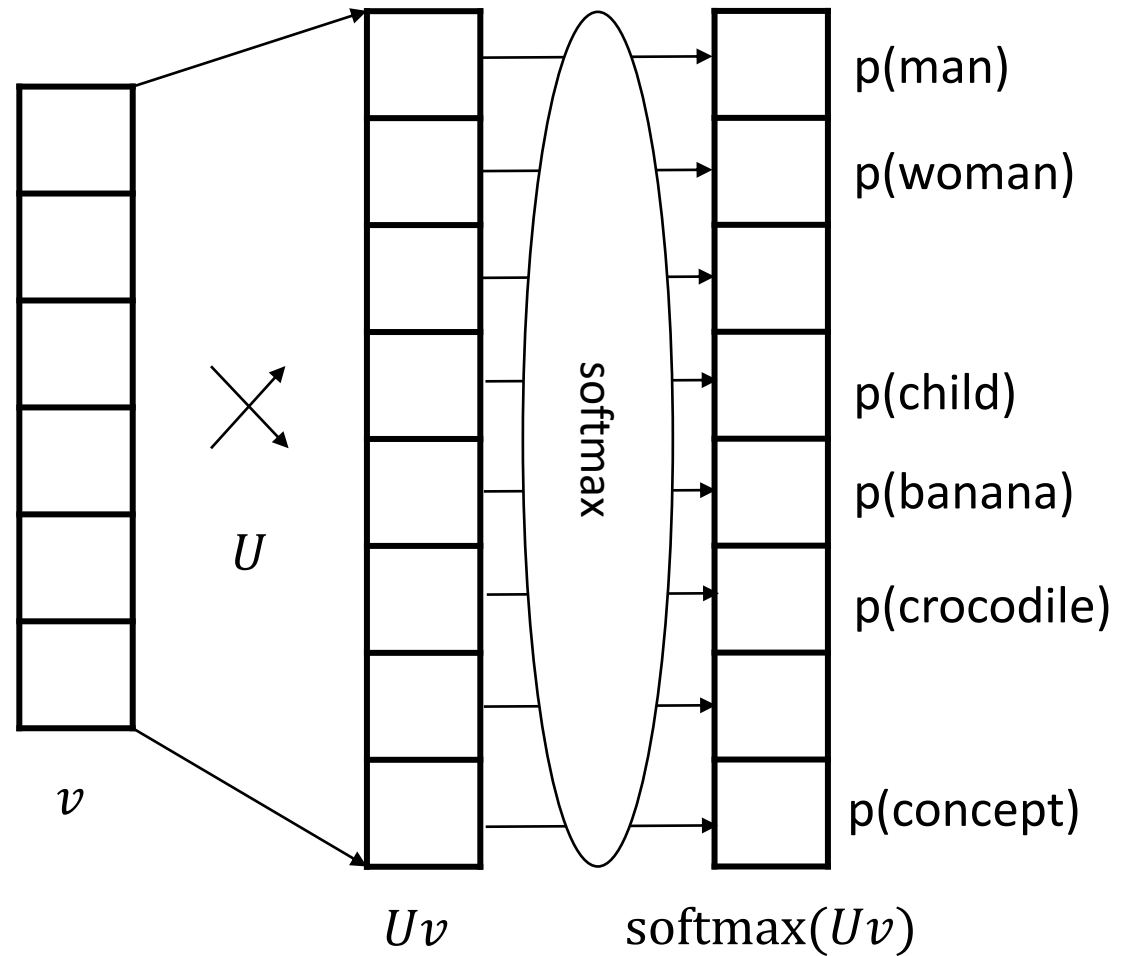
Our vector representation v
should be sufficient to
predict likely context words



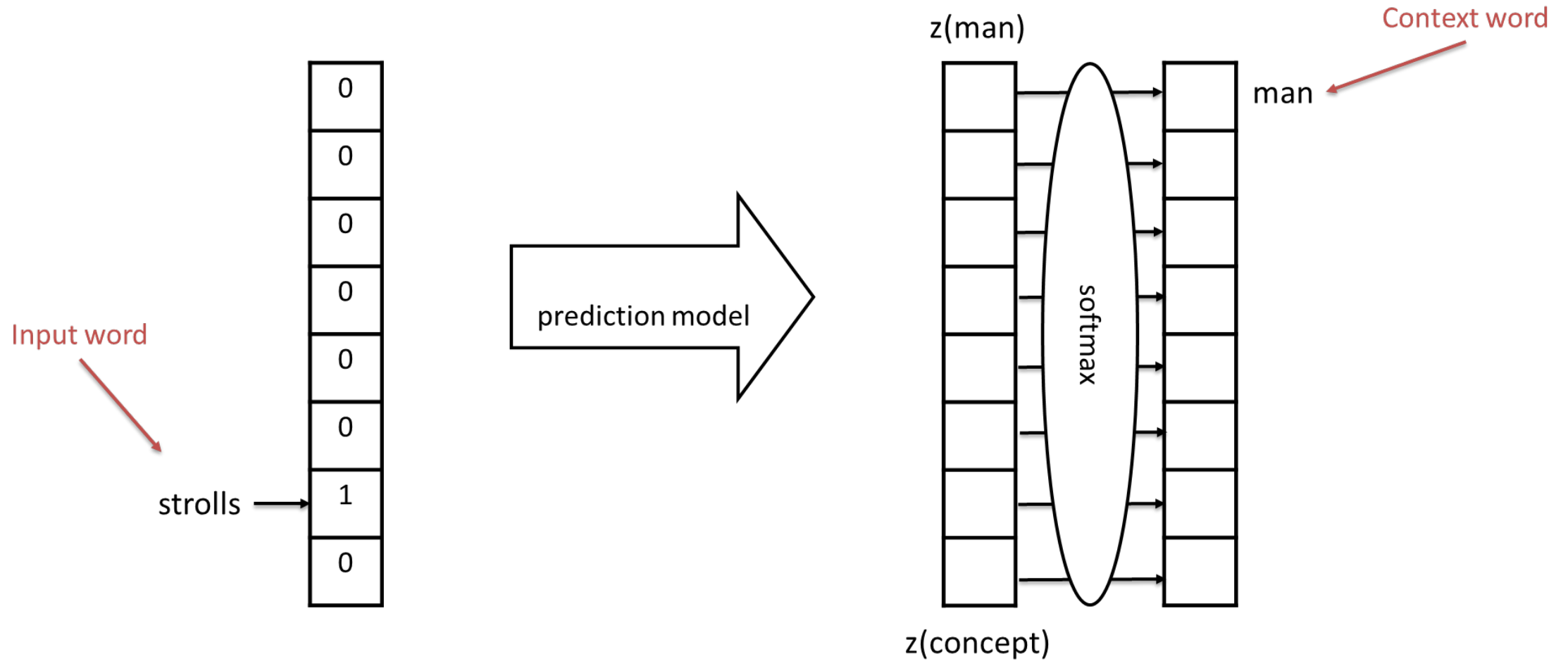
The skip-gram model

How do we *learn* these vectors?

First we'll consider from first principles how we might predict context words based on input words



The skip-gram model



Input words and context words are one-hot encoded
(similar to bag of words representation)

The skip-gram model

Training Data:

HUGE number of pairs of the following form:

{input word, context word}

e.g. from Wikipedia

Examples:

{strolls, man}

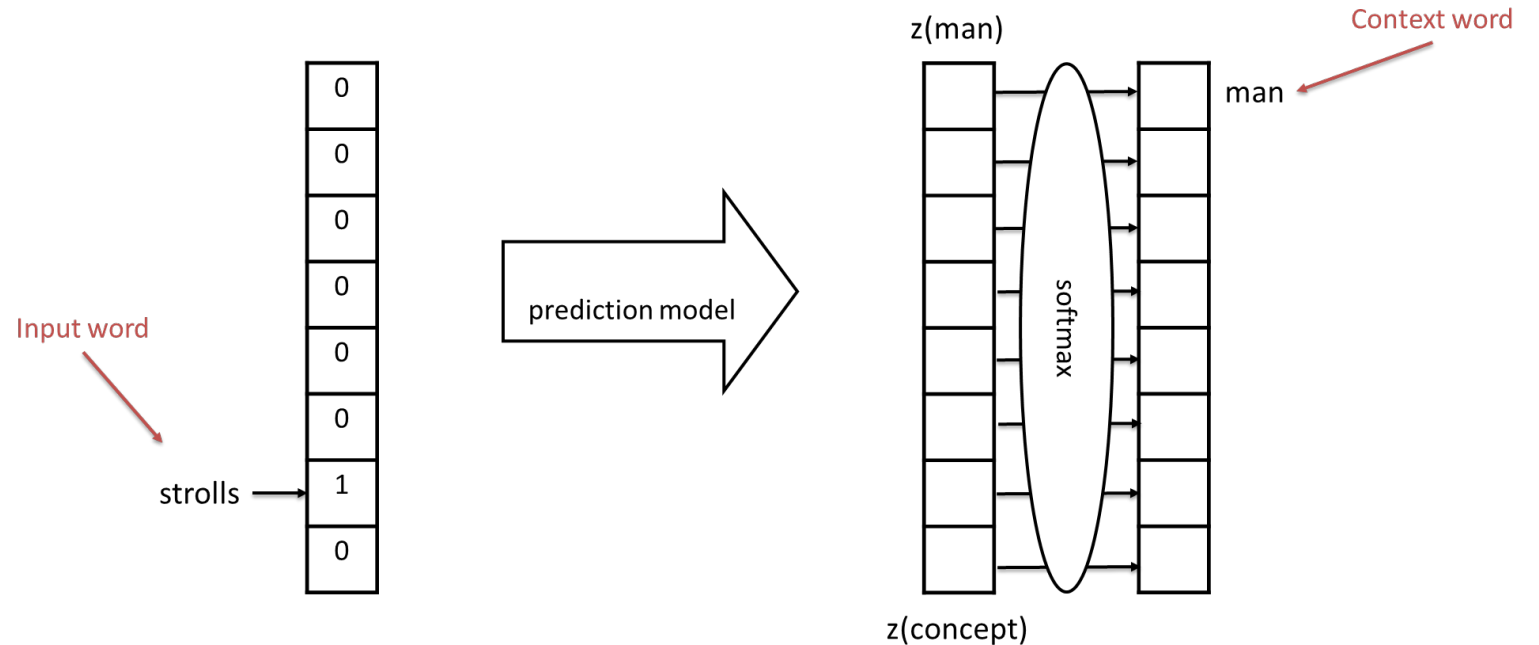
{strolls, woman}

{swims, crocodile}

{swims, fish}

{flies, bird}

{flies, plane}



The skip-gram model

Training Data:

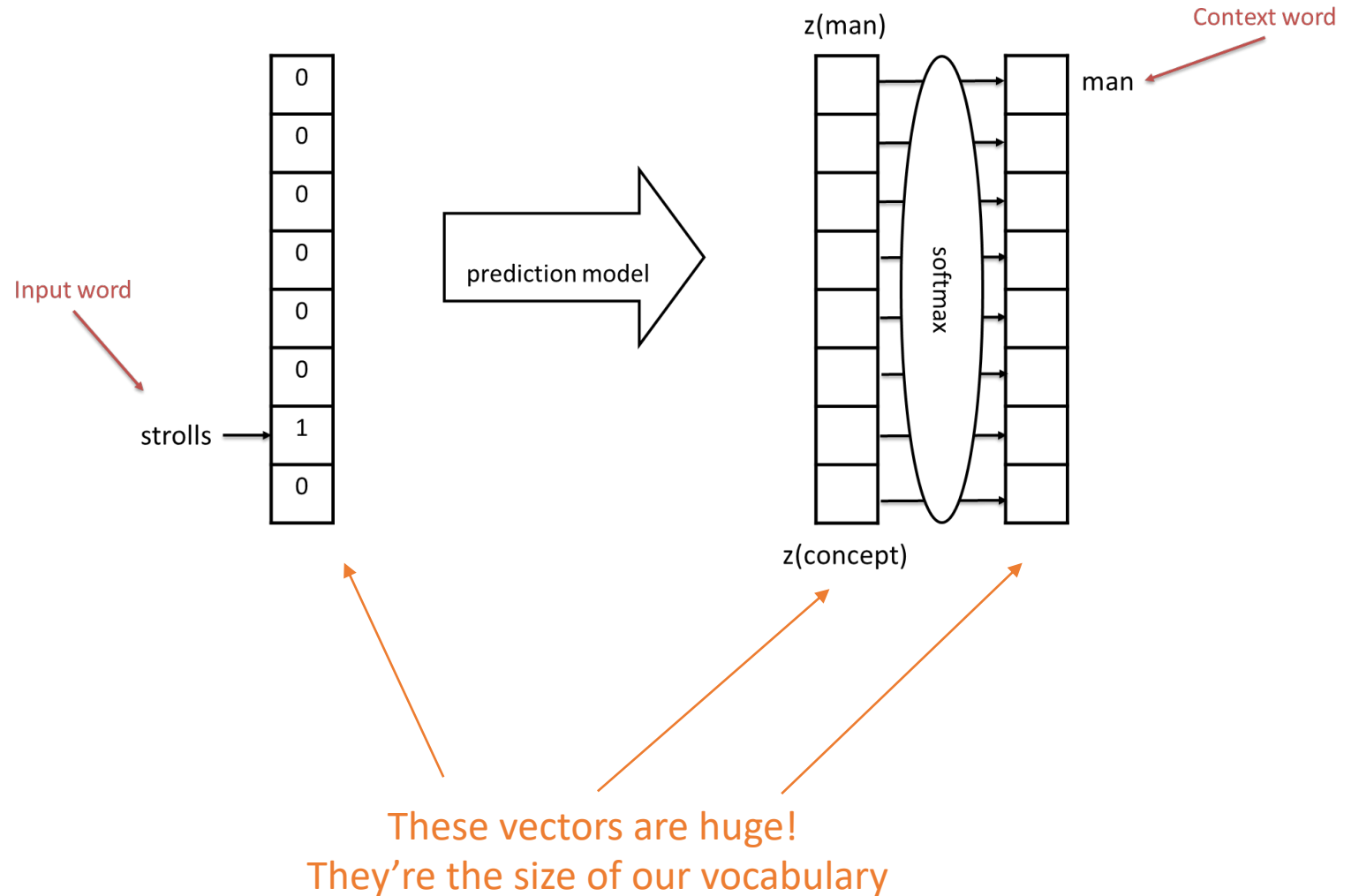
HUGE number of pairs of the following form:

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Examples:

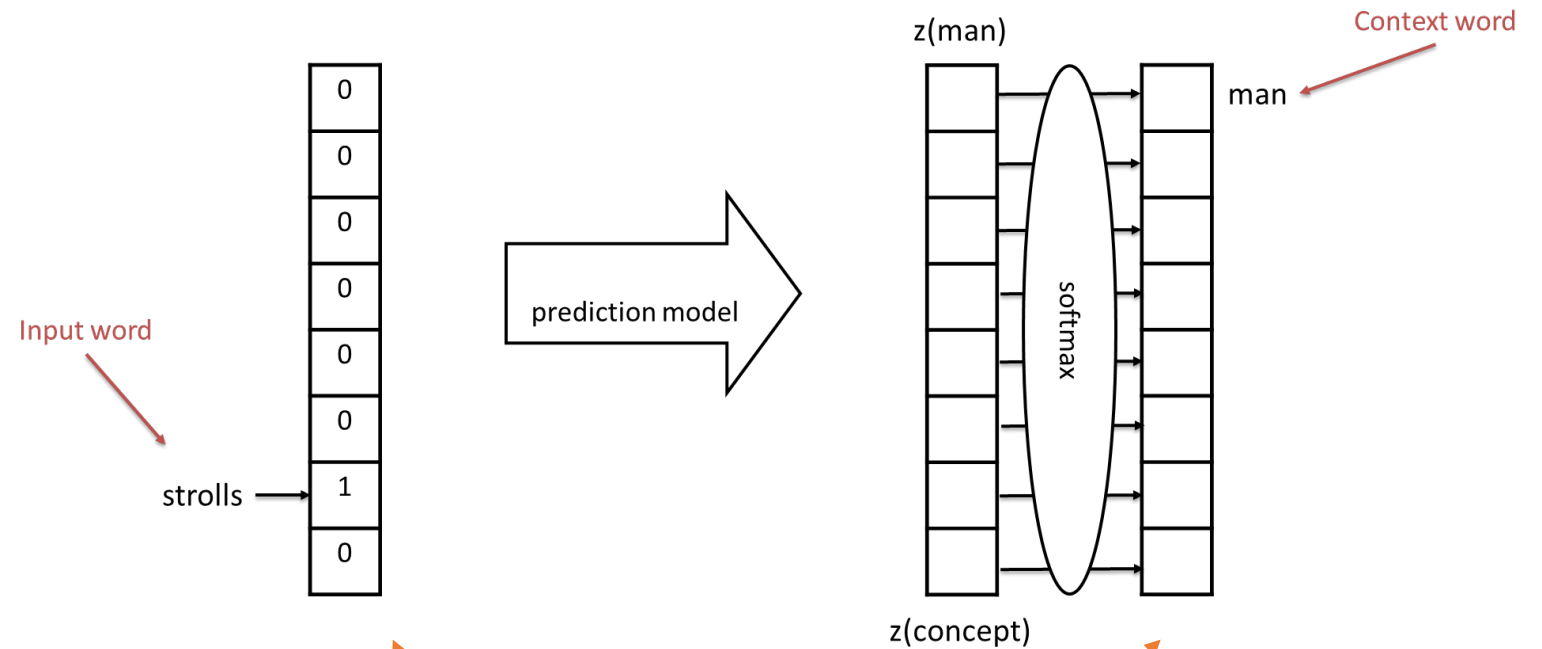
{strolls, man}
{strolls, woman}
{swims, crocodile}
{swims, fish}
{flies, bird}
{flies, plane}



The skip-gram model

First idea:

Directly connect our input
to the log-odds layer



These vectors are huge!
They're the size of our vocabulary

The skip-gram model

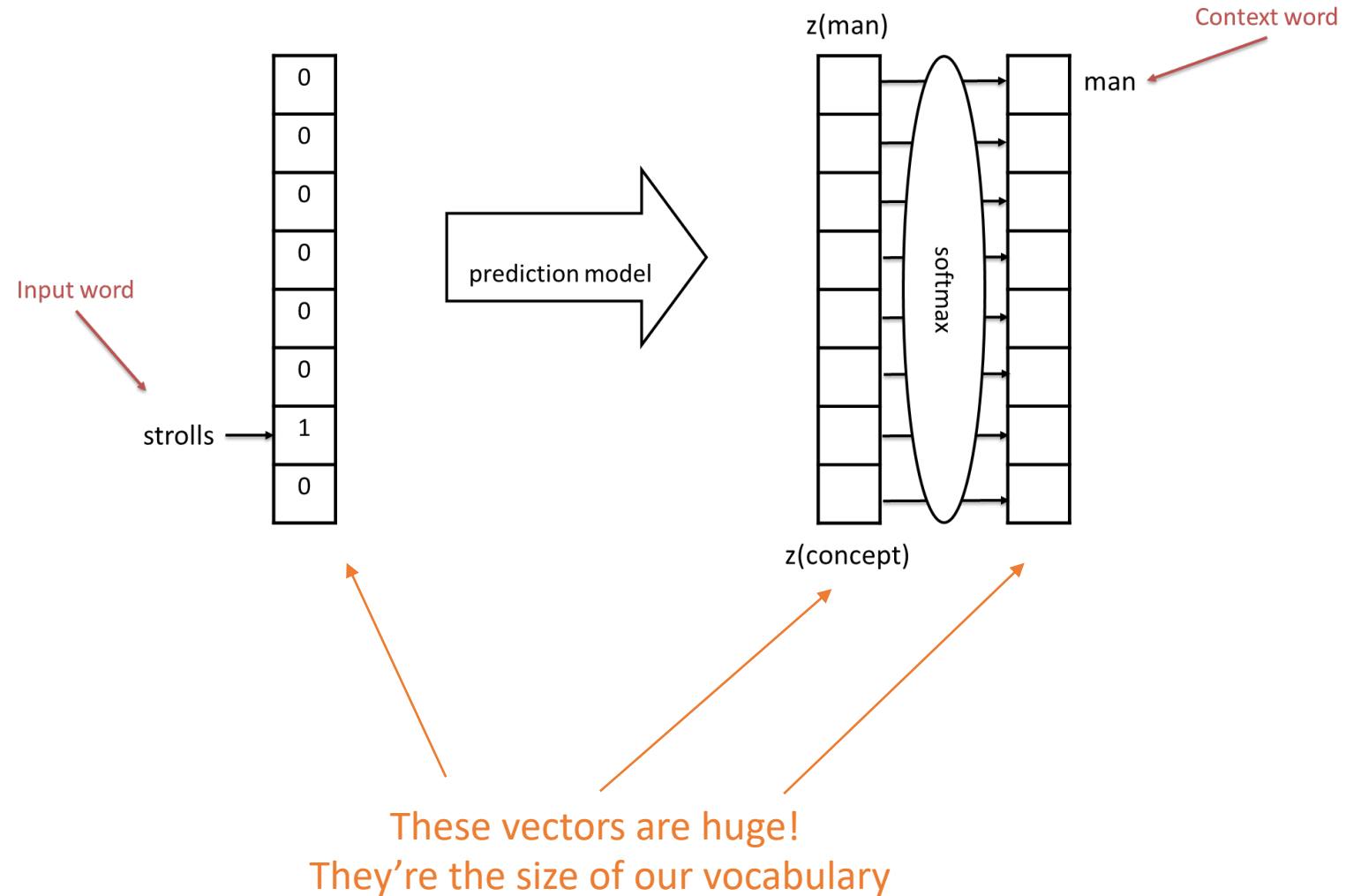
First idea:

Directly connect our input
to the log-odds layer

How many connections?

$V \times V$

Where V is our
vocabulary size
(approx. 50k)



The skip-gram model

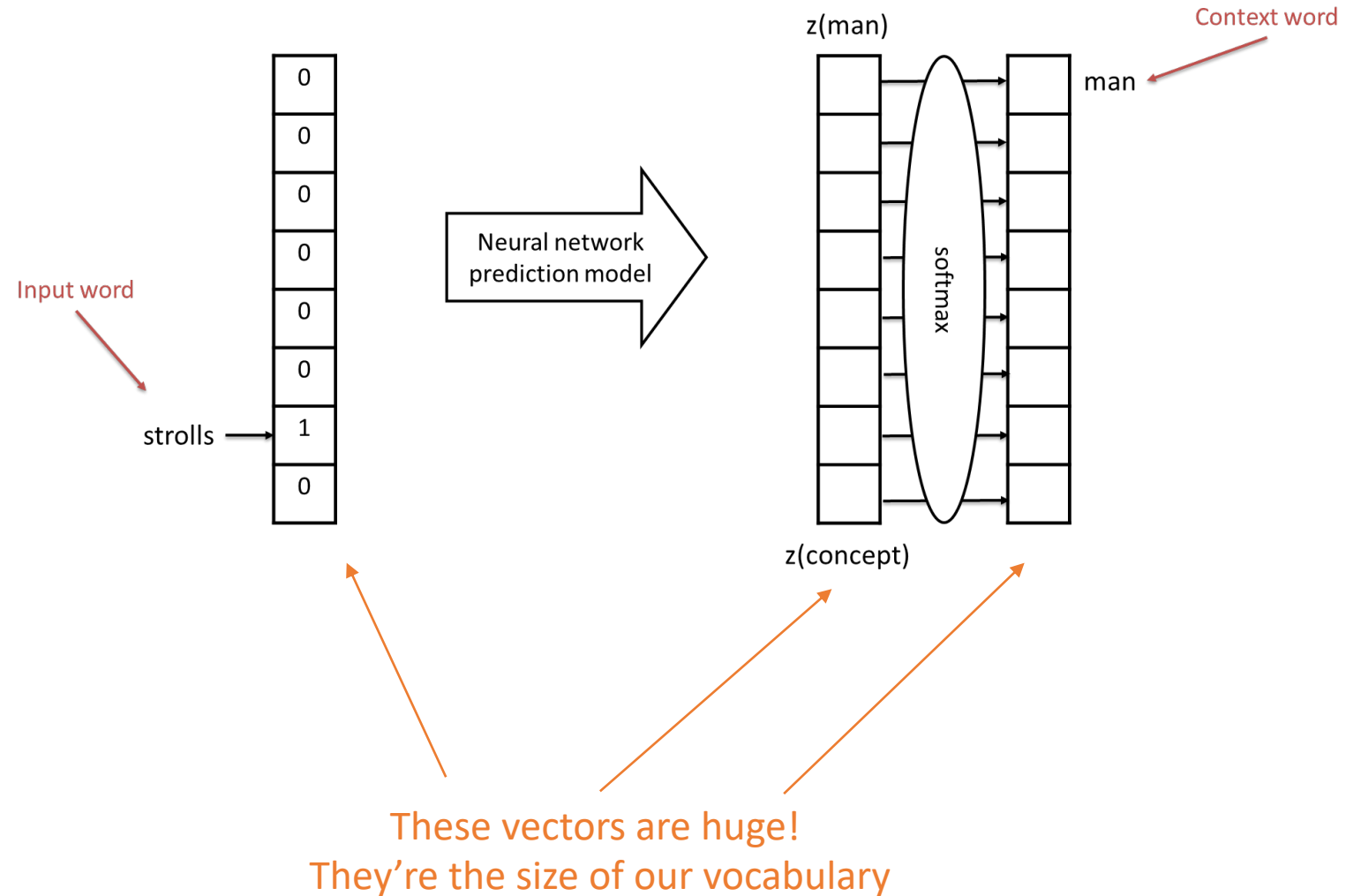
How about a single hidden layer?

How many connections?

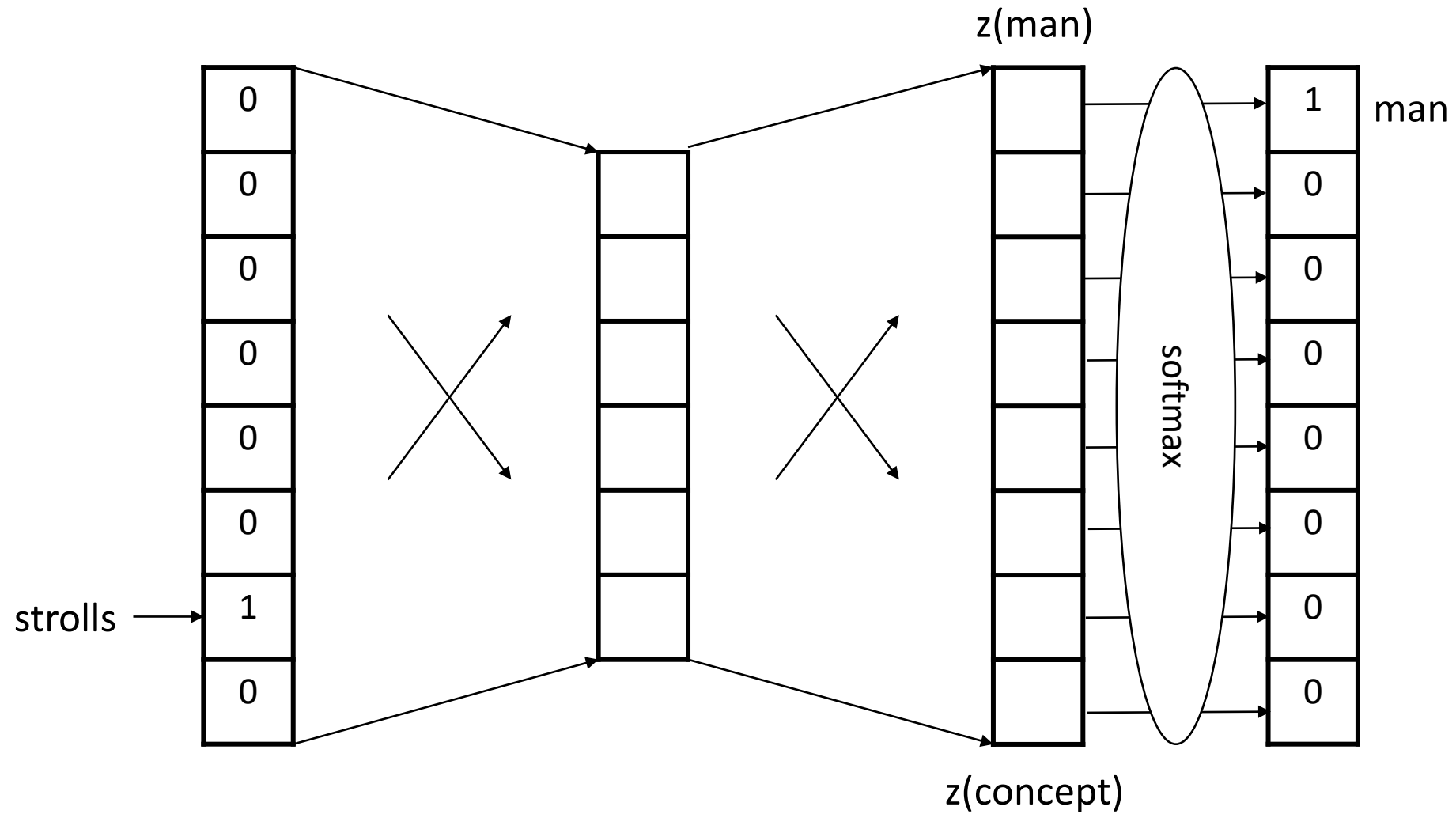
$V \times H \times 2$

Where V is our vocabulary size (approx. 50k)

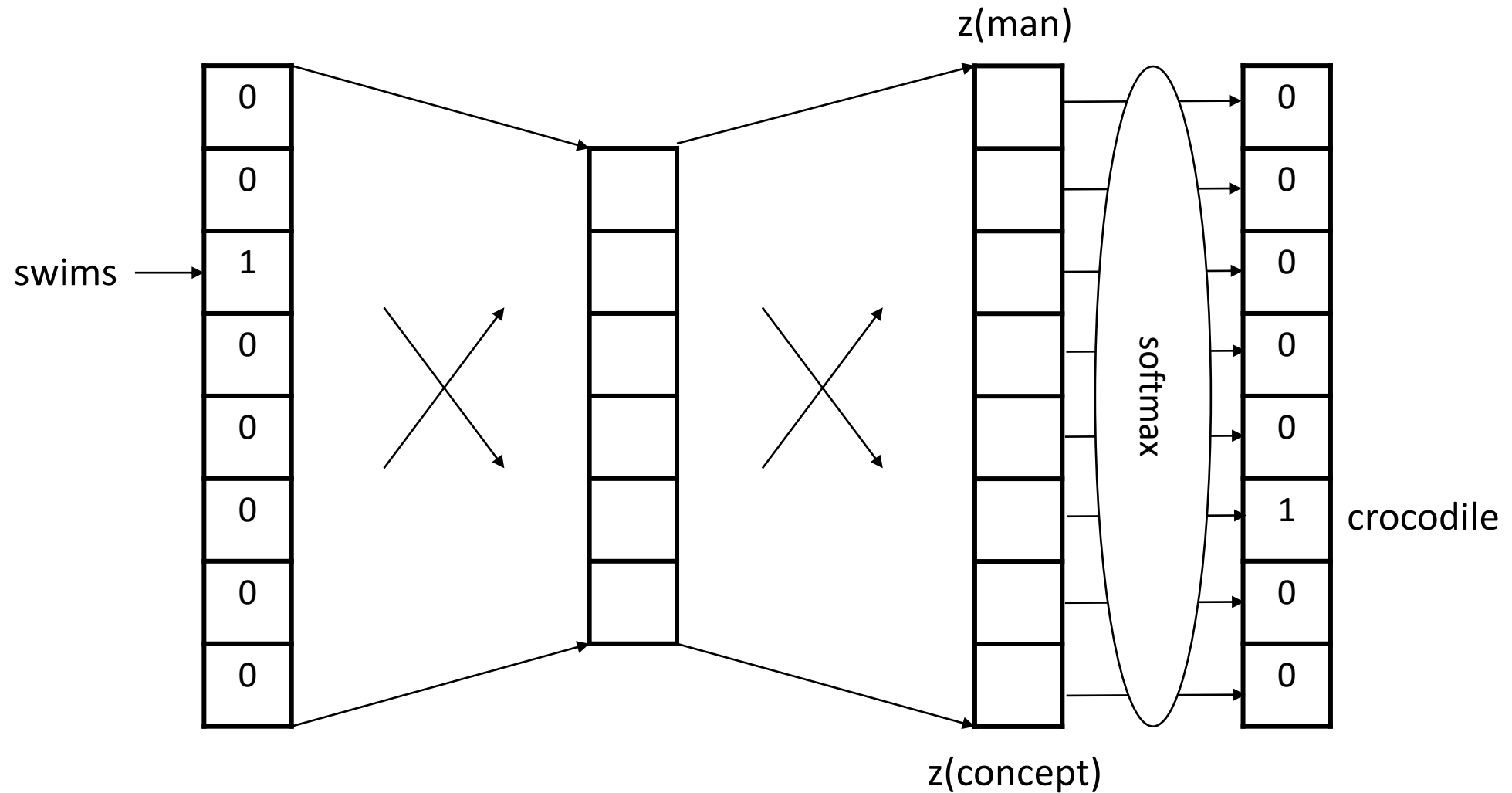
And H is our hidden layer size ($\ll V$)



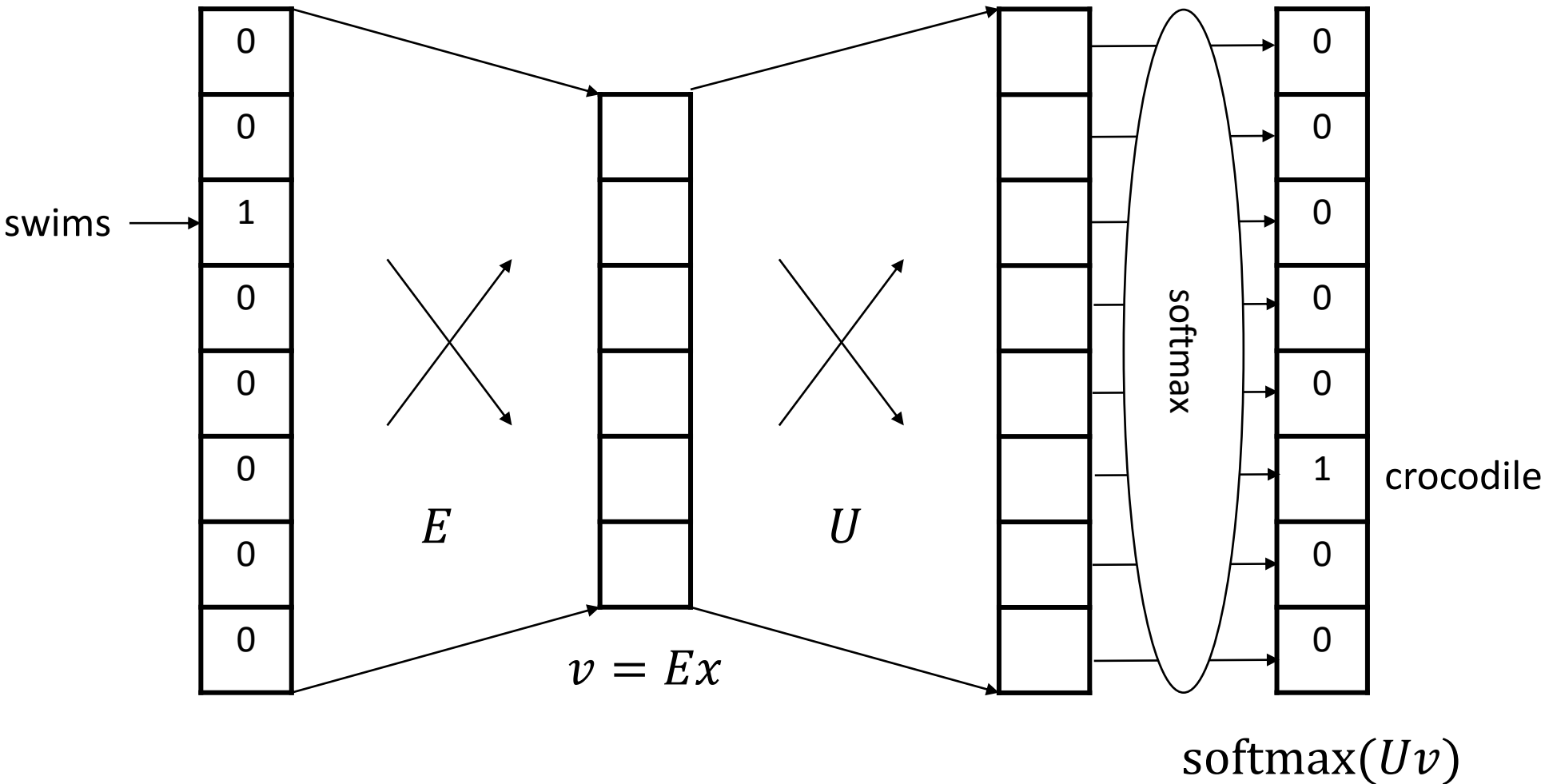
The skip-gram model



Use mini-batches of training examples; minimize cross-entropy loss

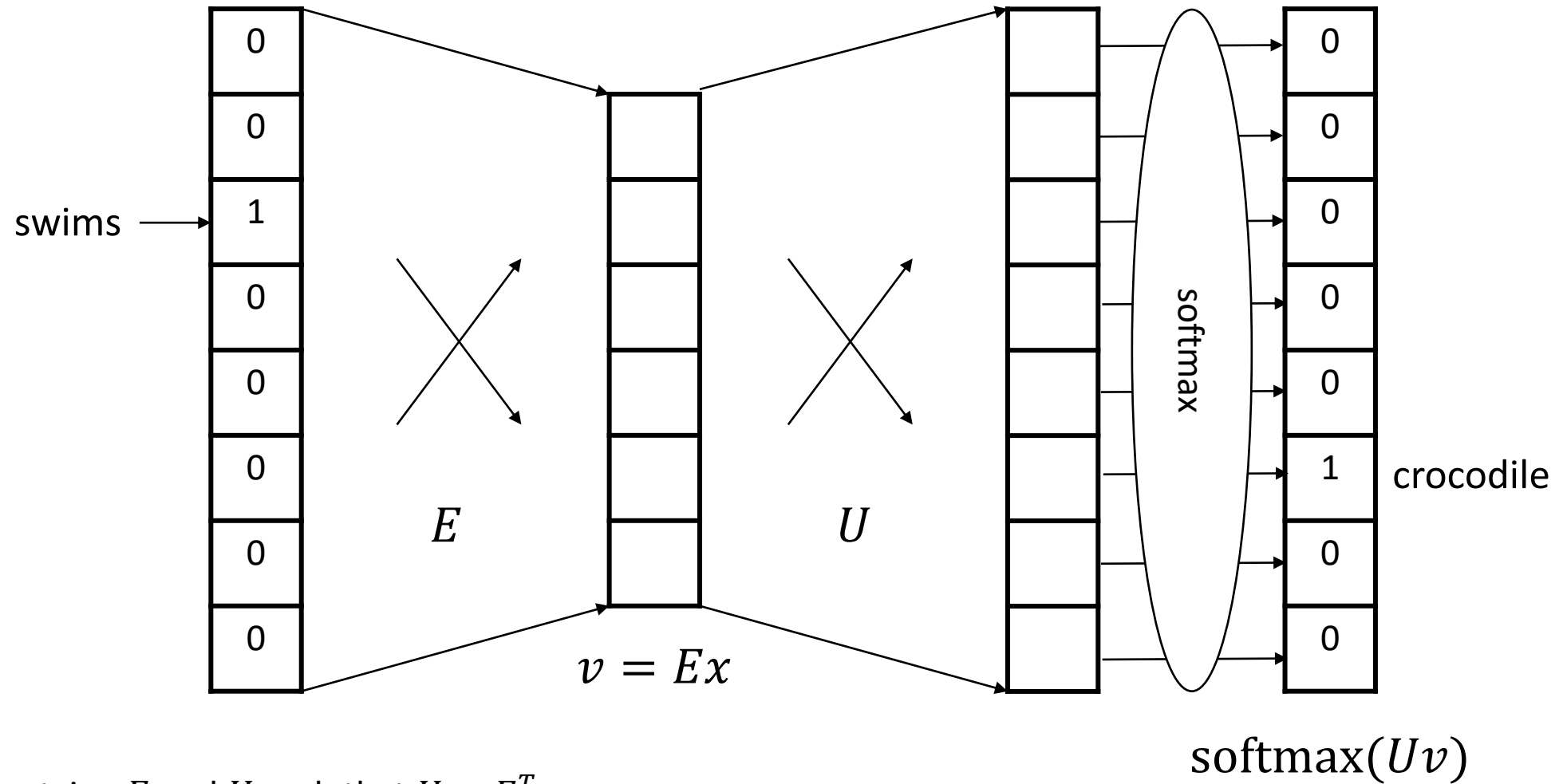


Weight Matrices E and U
(input embeddings E ; target embeddings U)



Weight Matrices E and U

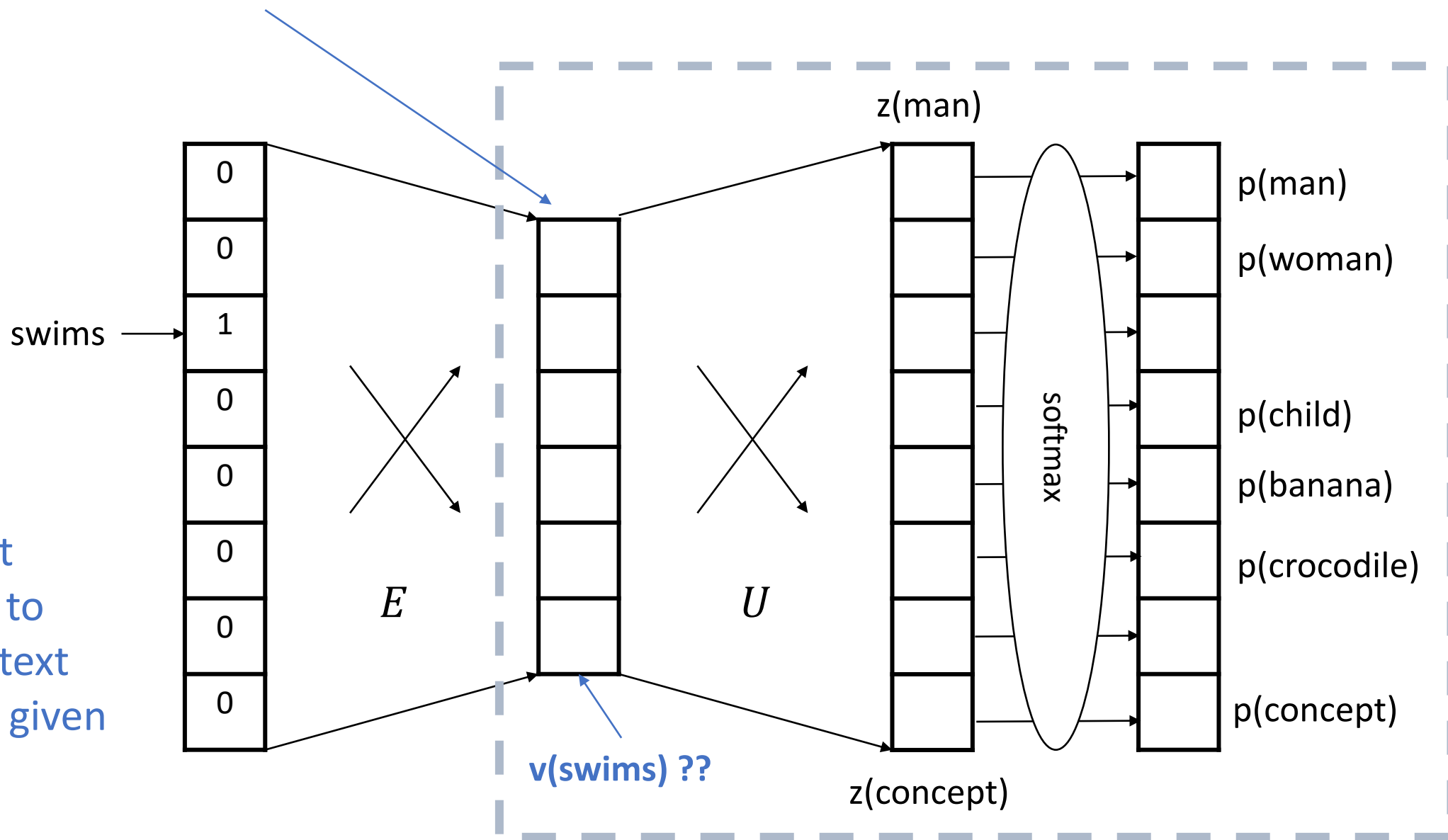
(input embeddings E ; target embeddings U)



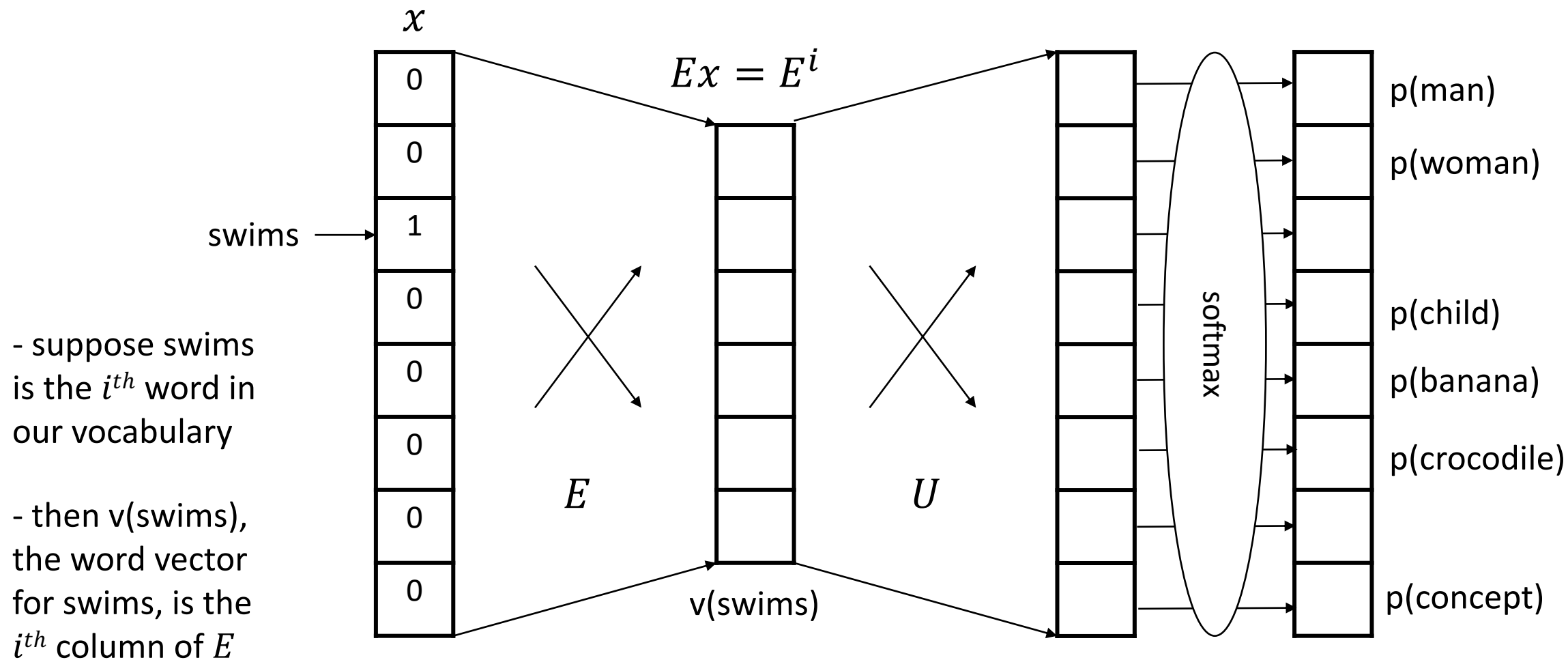
*consider tying E and U such that $U = E^T$

This is the vector we need; the embedding of *swims*

- it's compact
- It allows us to predict context words for a given input word

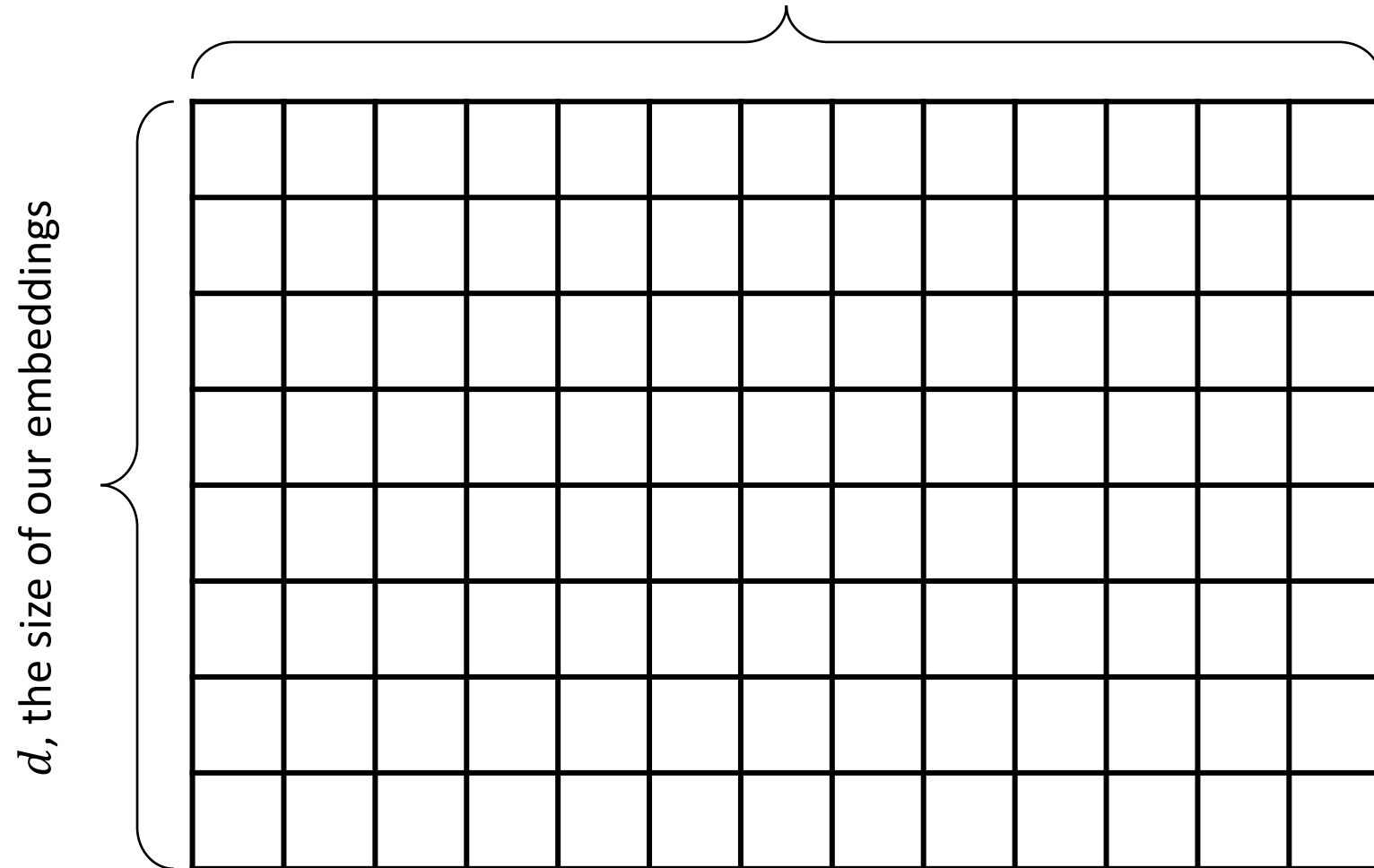


E is a look-up table of word embeddings

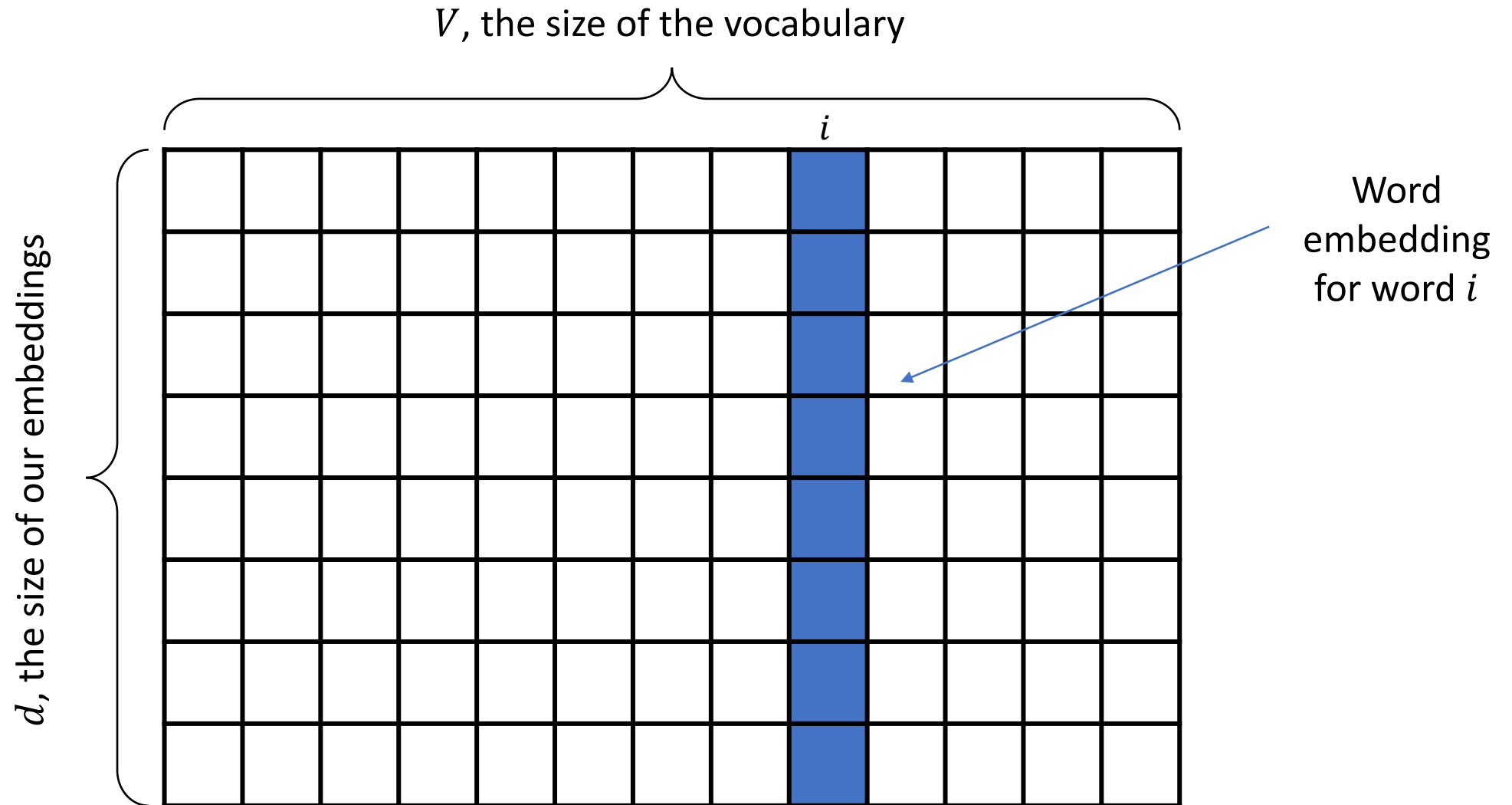


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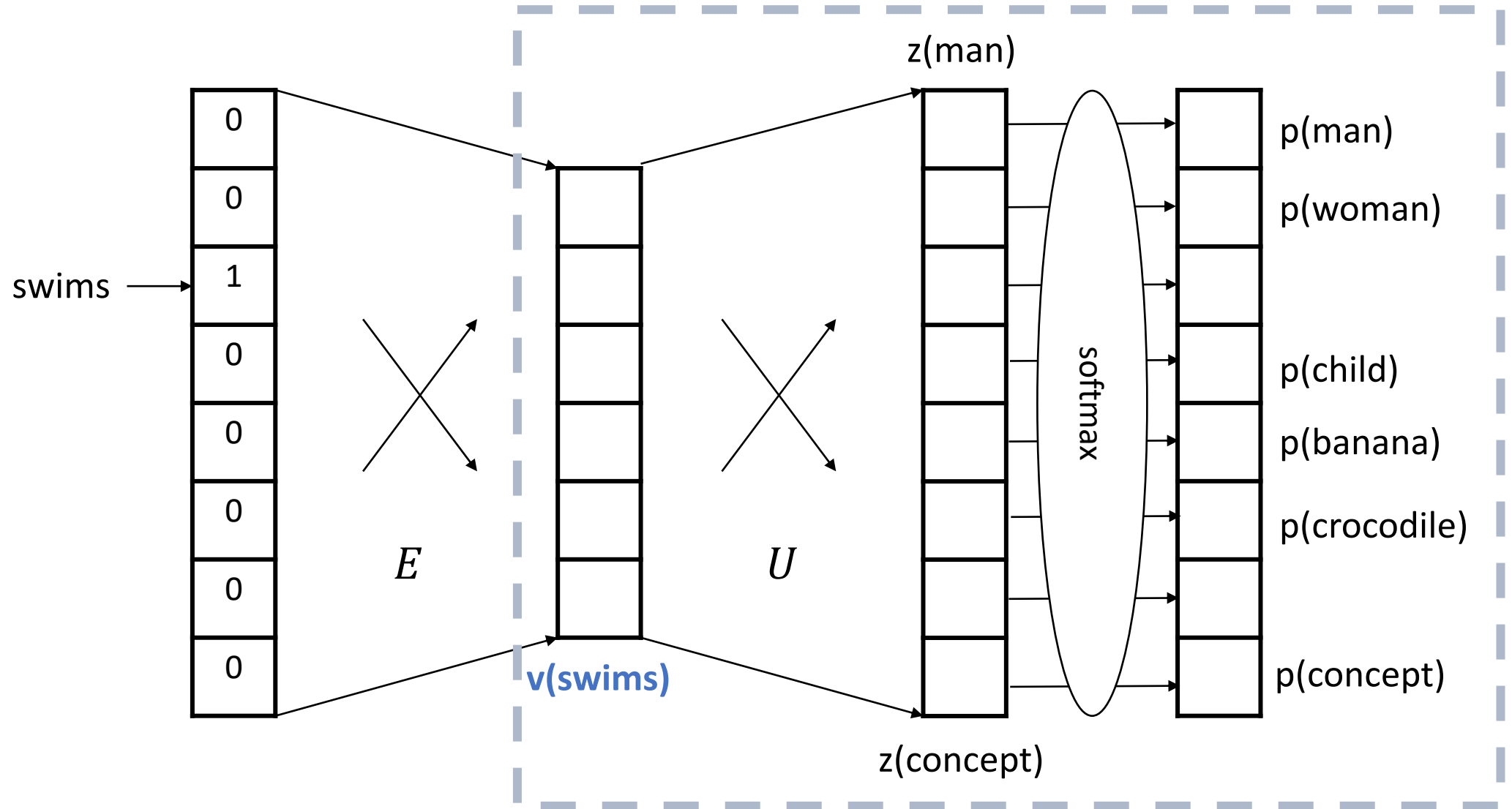
V , the size of the vocabulary



E is a look-up table of word embeddings



We now have a distributed representation of word *meaning* based on *context*



Variation 1: Continuous bag of words (CBOW)

Continuous Bag of Words (Mikolov et al., 2013)

Goal: Given surrounding context words, predict the center word

Architecture:

context words → look up embeddings → average → predict center word

Example: “A man [?] down the street” → predict “strolls”

$v(\text{center}) \approx \text{average}(v(A), v(\text{man}), v(\text{down}), v(\text{the}), v(\text{street}))$

Properties: Faster to train than skip-gram; works well for frequent words

Variation 2: negative sampling

Skip-gram with Negative Sampling (Mikolov et al., 2013)

Goal: Given a center word, predict which context words are real vs. noise

Training:

For each real pair (target, context), sample k “noise” words at random

Binary classification: is this a real pair or a noise pair?

Prediction: $p(\text{real}) = \sigma(\mathbf{v}_{\text{target}} \cdot \mathbf{v}_{\text{context}})$

Key advantage: Only need k+1 outputs per example (not full V-word softmax)

Works better than CBOW for rare words and larger datasets

Variation 3: factor the co-occurrence matrix

Step 1: Build a co-occurrence matrix M

M is $V \times V$, where $M[i,j]$ = count of times word i appears near word j in the corpus

Step 2: Factor using truncated SVD

$$M \approx U_d \Sigma_d V_d^T$$

Keep only the top d singular values ($d \ll V$)

Step 3: Extract word vectors

Word vector for word i = row i of U_d (or $U_d \Sigma_d$)

A compact d -dimensional representation that captures co-occurrence structure

Important detail: embeddings have norm 1

Constraint: $\|v\| \leq 1$ (often normalized so $\|v\| = 1$)

Why constrain the norm?

- Without constraint, embeddings can grow arbitrarily large during training
- Large norms dominate similarity calculations regardless of direction
- We want similarity to reflect meaning (direction), not magnitude

Key benefit:

With unit norm, **cosine similarity = dot product**

$$\cos(a, b) = a \cdot b / (\|a\| \cdot \|b\|) = a \cdot b \quad (\text{when } \|a\| = \|b\| = 1)$$

All embeddings lie on the surface of a unit sphere in d-dimensional space

What can we do with word embeddings...

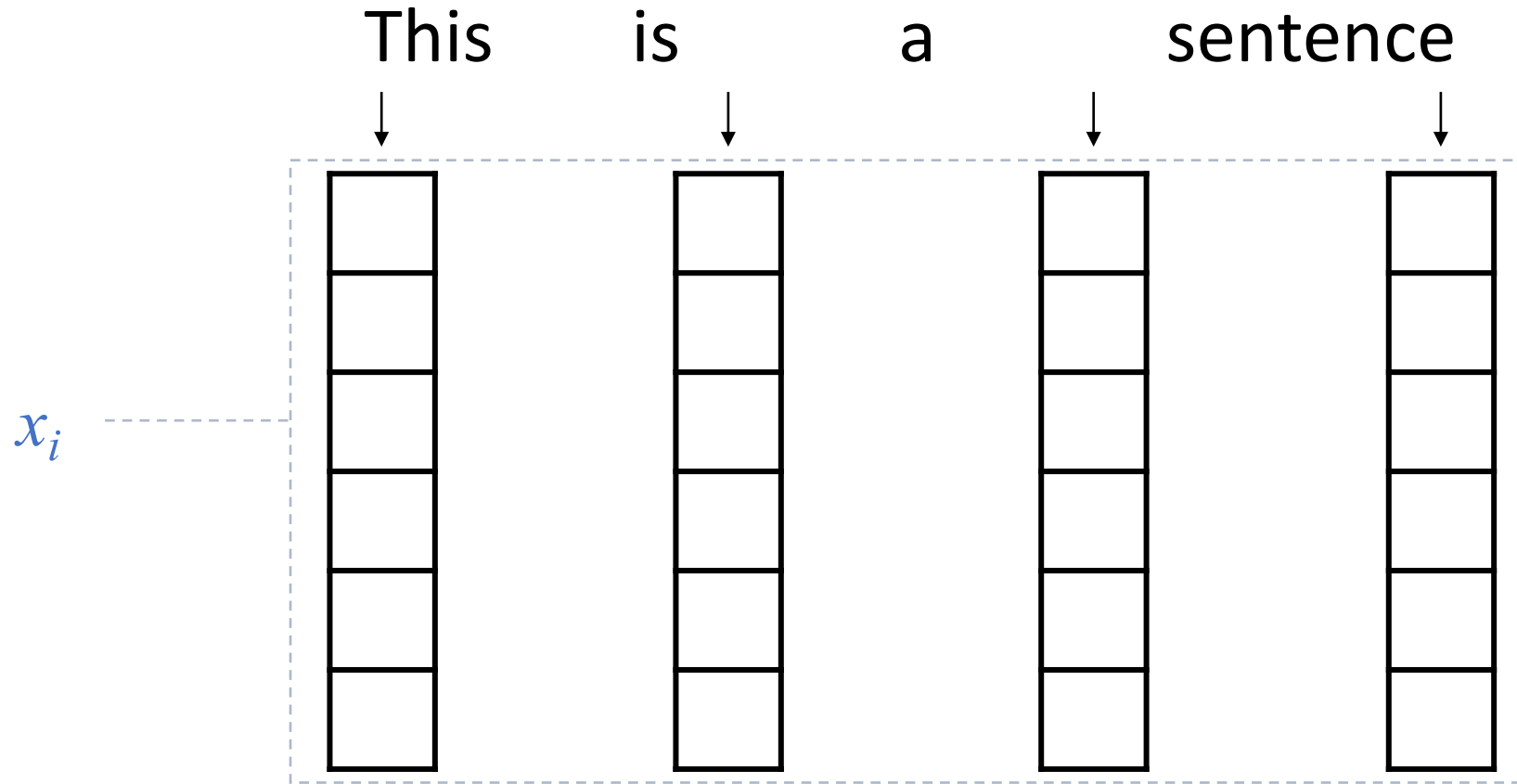
...to learn to extract increasingly complex aspects of meaning

Useful feature 1: retrieval

- We wish to use embeddings to find words that have similar meaning to a given word
- Step 1: obtain the embedding for a word of interest
- Step 2: calculate cosine similarity with all other words in the vocabulary
- Consider: why can't we do retrieval with our one-hot representations?

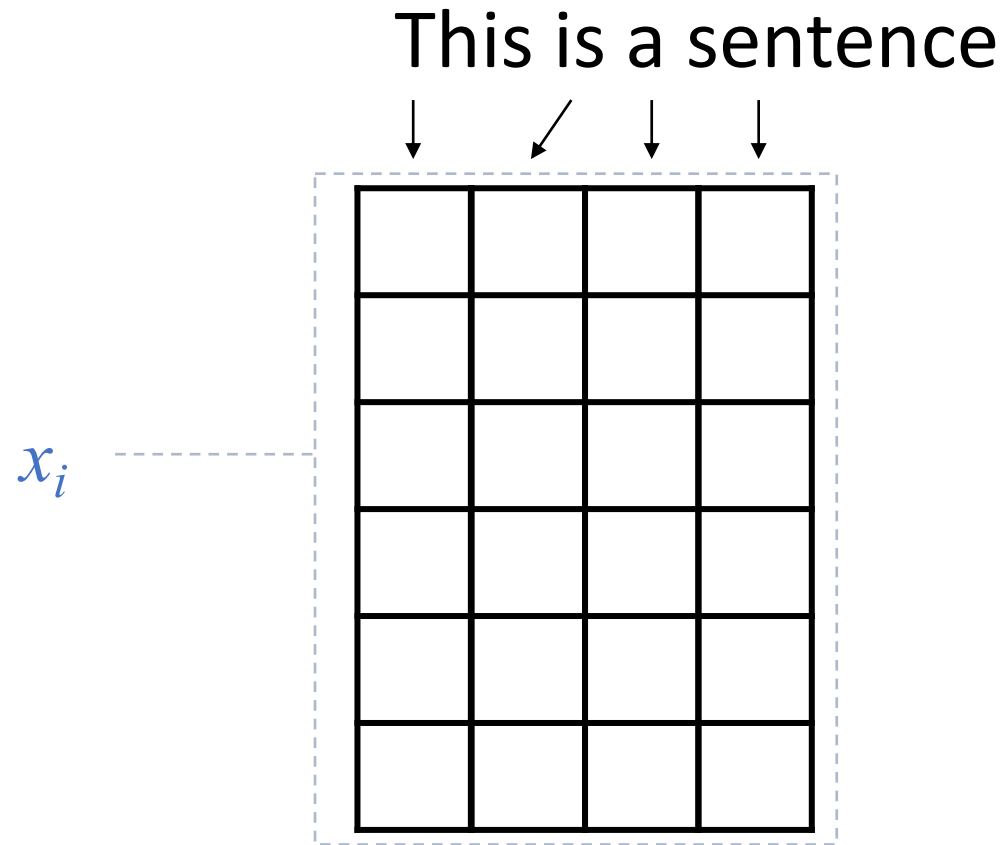
Useful feature 2: applying existing models

- Look up words individually to obtain their vectors
- Construct a sequence of vectors



What happens when we embed all words in a sentence?

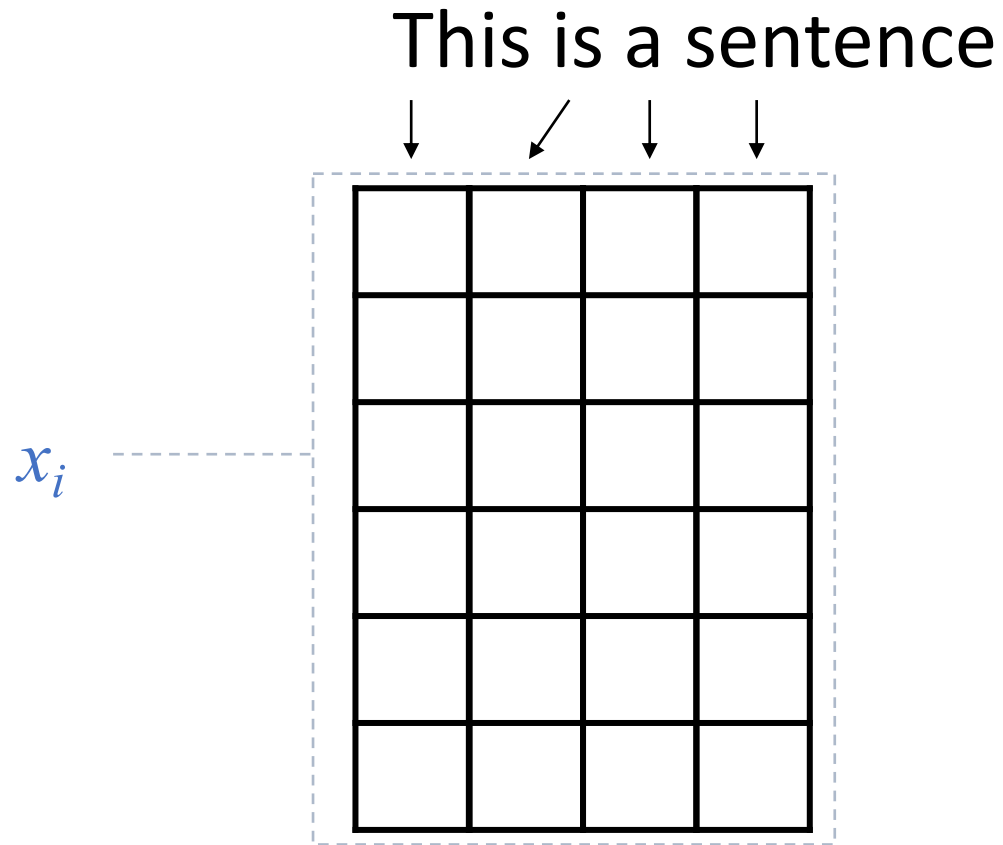
- Look up words individually to obtain their vectors
- Construct a sequence of vectors



Now we have a grid of numbers
Similar in many ways to an image

What happens when we embed all words in a sentence?

- Look up words individually to obtain their vectors
- Construct a sequence of vectors



How do we aggregate over the sequence length?

- Mean pooling
- Max pooling
- *Learned* pooling (i.e. linear attention)
(we will return to this next time)

Now we can use deep learning to build our hierarchy of features.



pixels

low-level
motifs

- edges
- shapes
- textures

high-level
motifs

- eyes
- ears
- paws

dog

label

End goal: predict *dog* from *pixels*

Now we can use deep learning to build our hierarchy of *semantic* features.

Chief Complaint:
Shortness of breath.

History of the Present Illness:
Mr. ■ is a previously healthy 56-year-old gentleman who presents with a four day history of shortness of breath, hemoptysis, and right-sided chest pain. He works as a truck driver, and the symptoms began four days prior to admission, while he was in Jackson, MS. He drove from Jackson to Abilene, TX, the day after the symptoms began, where worsening of his dyspnea and pain prompted him to go to the emergency room. There, he was diagnosed with pneumonia and placed on Levaquin 500 mg daily and Benzonatate 200 mg TID, which he has been taking for two days with only slight improvement. He then drove from Abilene back to Greensboro, where he resides, and continued to experience shortness of breath, right sided chest pain, and hemoptysis. He presented to an urgent care office in town today, and was subsequently transferred to the Moses Cone ER due to the provider's suspicion of PE.

The right-sided pain is located midway down his ribcage, below the axilla. This pain is sharp, about 7/10 in severity, and worsens with movement and cough. Pressing on the chest does not recreate the pain. He feels that the pain has improved somewhat over the past two days. The hemoptysis has been unchanged since it began; there is not frank blood, but his sputum has been consistently blood-tinged. The blood seems redder at night. The dyspnea has been severe, and it is difficult for him to walk more than across a room. He states that he feels as though there is a "rattling" in his chest. At baseline, he experiences no dyspnea on exertion and has no history of COPD or other respiratory problem. He is a smoker, smoking a little less than a pack a day for thirty-five years. Past history is notable for the fact that he experienced transient left lower leg swelling – from below the knee down – and pain several weeks ago during a cross-country haul. He also notes a four day history of decreased appetite, poor sleep, and subjective fever and chills, with a measured fever of 103 in the hospital in Abilene. He had a bout of pneumonia about two months ago, but has been healthy for the most part and denies any chronic medical conditions. Currently he is fairly comfortable, with morphine helping with the pain. He has no history of a clotting disorder, no cardiac history, and denies any chest trauma or aspiration. He has had no sick contacts.

Past Medical History:
1. Hernia repair
2. Bilateral thumb surgeries, secondary to two separate injuries sustained while working with machinery

Medications:
No regular medications, over-the-counter medications, or supplements. Has taken two days of the medications prescribed by the ER in Abilene: Levaquin 500 mg daily and Benzonatate 200 mg TID.

grid of
semantic
attributes

low-level
motifs

- words
- short phrases

high-level
motifs

- concepts
- topics

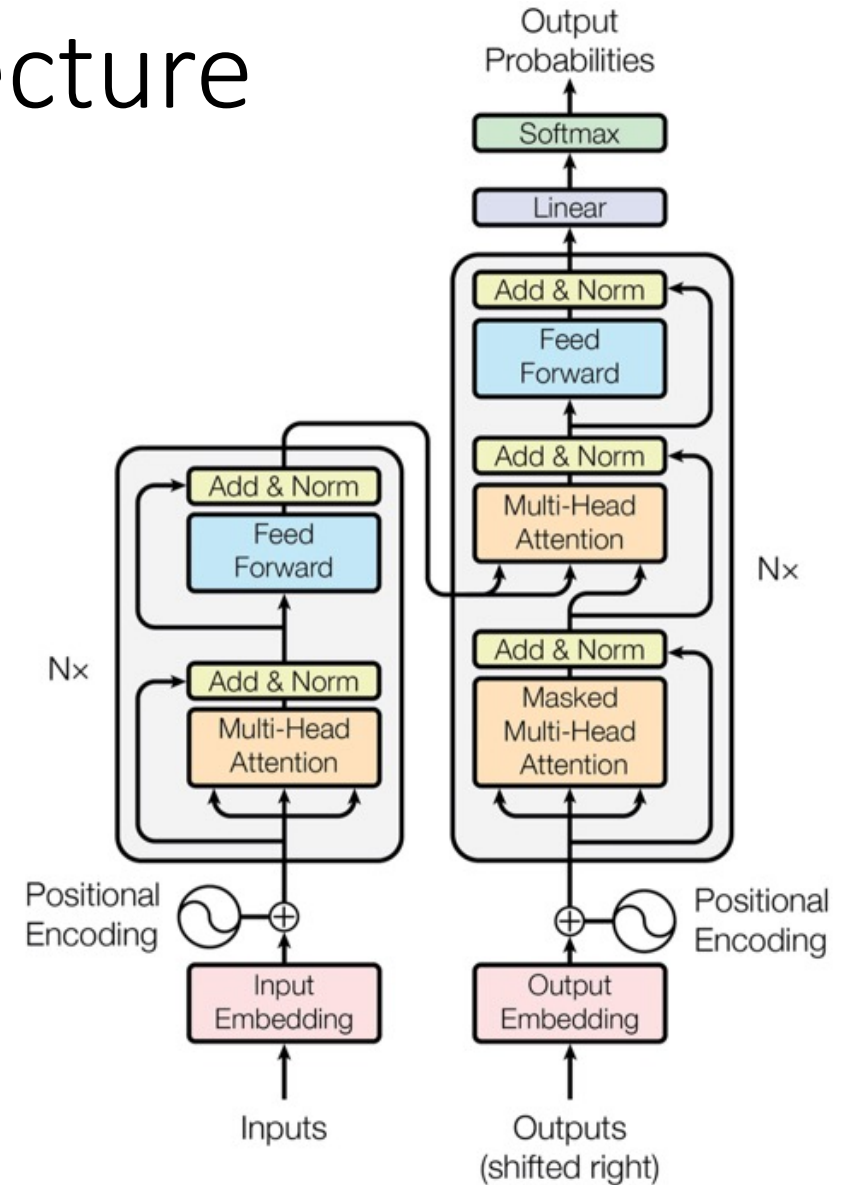
PE

label

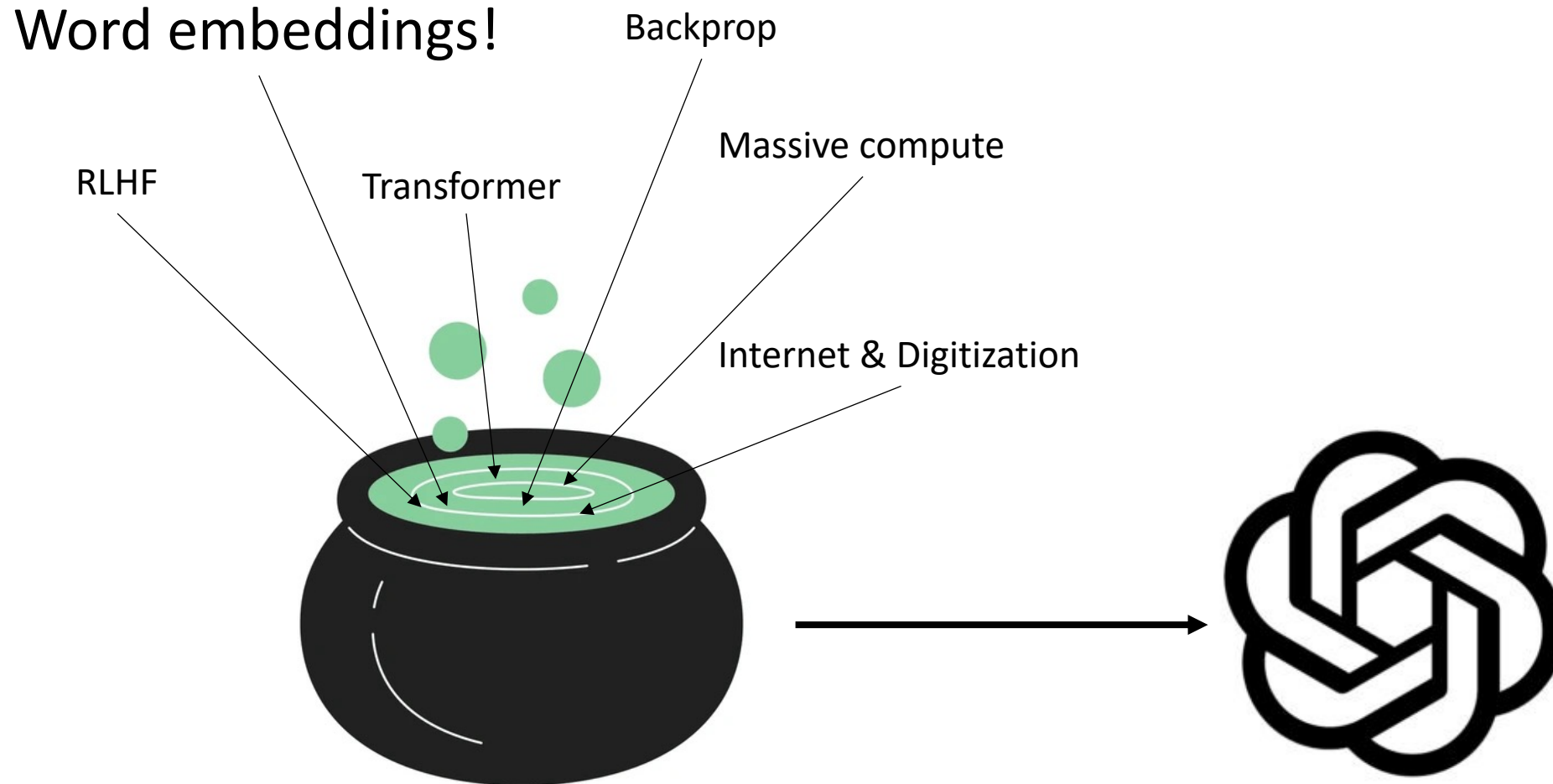
End goal: predict *pulmonary embolism* from *text*

Next time: Transformer architecture

- In current LLMs, embeddings are learned end to end with the rest of the model
- However, the embedding concept remains central
- We may view this architecture as *progressively refining our embeddings based on context*



Recipe for ChatGPT / modern LLMs



Important Takeaways:

- We are learning a vector representation for each word based on the contexts in which it appears
- training data: large number of pairs of nearby words from a large corpus
- These vectors give us much more flexibility when modeling: makes text sequences like other sequences

Medical concept embeddings

Recall: our first step was to define the self-supervision task

{input word, context word}

{strolls, man}

{strolls, woman}

{swims, crocodile}

{swims, fish}

{flies, bird}

{flies, plane}

We define a context word as one that appears inside a fixed-length window around the input word in our training corpus.

(e.g. Wikipedia)

A man strolls down the street.

input context

Mini activity: how should we define the self-supervision task for medical codes?

- If we know how to define {input code, context code} pairs, we can use any of the approaches we've described
 - Skip gram
 - CBOW
 - Negative sampling
 - Matrix factorization
- How should we define these pairs?